

PYTHON-TEST-03 MAY-B

Section A: MCQs

1M× 10 = 10 M

1. Which data structure is mutable in Python?
 - a) List
 - b) Tuple
 - c) String
 - d) All of the above
2. What will be the output of `len({1, 2, 2, 3})`?
 - a) 4
 - b) 3
 - c) 2
 - d) Error
3. Which method adds an item to a list?
 - a) `insert()`
 - b) `append()`
 - c) `extend()`
 - d) `add()`
4. What does `dict.get("key")` return if the key does not exist?
 - a) Error
 - b) 0
 - c) None
 - d) False
5. Which of the following is a key difference between a set and a list?
 - a) Sets are ordered
 - b) Lists are immutable
 - c) Sets do not allow duplicates
 - d) Lists do not allow duplicates
6. What is the output of `type({})`?
 - a) set

b) dict

c) list

d) tuple

7. Which method is used to remove a key-value pair from a dictionary?

a) del

b) pop()

c) remove()

d) discard()

8. Which of the following can be used as a dictionary key?

a) list

b) set

c) tuple

d) dict

9. What is the result of `set([1, 2, 2, 3, 4])`?

a) [1, 2, 3, 4]

b) {1, 2, 2, 3, 4}

c) {1, 2, 3, 4}

d) Error

10. Which of these is immutable?

a) List

b) Set

c) Tuple

d) Dictionary

Section B: 2-Mark Questions

10 × 2 = 20 M

1. Define a list and give an example.

2. Differentiate between list and tuple.

- 3.What is the use of .update() in dictionaries?
- 4.How do you remove duplicates from a list using set?
- 5.What is a tuple? Write syntax and example.
- 6.Write a Python statement to check if a key exists in a dictionary.
- 7.What does the clear() method do in sets?
- 8.Define a nested dictionary with at least two keys.
- 9.How do you convert a list to a tuple?
- 10.Explain the use of union() and intersection() in sets.

Section C: 5-Mark Questions

4 × 5 = 20M

1.Write a Python program to:

Create a list of 5 fruits

Add a fruit

Remove one

Sort the list

Print the final list

2.Write a Python program that:

Takes a sentence as input

Counts the frequency of each word

Stores it in a dictionary

Displays the result

3.Explain any 5 dictionary methods with examples.

4.Write a Python program that:

Creates two sets of numbers

Performs union, intersection, difference

Prints all the results